

ZACHARY GERBI

github.com/zgerbi | zachary.gerbi@gmail.com | (321) 591-4790

EMPLOYMENT & LEADERSHIP EXPERIENCE

Universal Avionics – Software Engineer I

September 2023 – June 2026
Duluth, GA

- Developed and implemented embedded C++ software for safety-critical avionics systems.
- Delivered full-stack development on connected avionics apps for iPad, powering data-driven fleet and post-flight telemetry analysis with AWS.
- Collaborated with systems and hardware engineers to integrate new avionics components across diverse aircraft configurations.
- Enforced software development, verification, and documentation standards compliant with DO-178B.

Epsilon C5I – Software Engineer Intern

May 2022 – August 2022
Largo, FL

- Synchronized software deliverables on a \$280M defense contract using Python, C++, and Jenkins CI pipelines.
- Optimized source code storage strategies to reduce repository footprint by 38%.
- Collaborated within an Agile cross-functional team in an enterprise SAFe environment.
- Presented Sprint and Epic retrospectives to engineering leads and executive stakeholders.
- Held and maintained an active DOD Secret Clearance.

TECHNICAL SKILLS & ACCREDITATIONS

Languages: C++, C#, Java, Python, Swift, Kaitai, MATLAB, SQL

Cloud & Frameworks: Amazon Web Services (AWS Glue, AWS Lambda), React, PyTorch Lightning, JUCE

Embedded & Hardware: Quartus, Waveforms, DO-178B Standards

Developer Tools & CI/CD: Git, Jenkins, Unity, Linux, Windows, macOS, Atlassian Ecosystem

AI Tooling & Credentials: Ollama, GitHub Copilot, Gemini | SolidWorks Certified, DOD Secret Clearance (Held)

PROJECTS

UA FlightReview | Connected Avionics Flight Data Analysis | React, Swift, C#, AWS

- Ingests and visualizes FDR and FMS telemetry across fleet aircraft for pilots and maintenance crews to diagnose performance metrics and safety anomalies.
- Built with a React web interface backed by a modular Swift upload framework, processing telemetry via AWS Glue and Lambda.

GatorBones | AI/ML Image Segmentation | Python, PyTorch Lightning, CUDA

- Built a radiographic segmentation pipeline for UF's BRIO Lab to infer bone and joint coordinates from canine X-rays.
- Implemented a SwinUNETR transformer architecture in PyTorch Lightning augmented with a Boundary Patch Refinement layer, trained across 1,000+ scans on UF's HiPerGator supercomputer.

Soundstage | VST3 Spatial Audio Plugin | C++, JUCE

- Engineered a real-time 3D spatialization audio plugin applying Head-Related Transfer Functions (HRTF) from the CIPIC database.
- Implemented real-time dynamic audio placement responsive to MIDI inputs and on-screen controls.

EDUCATION

University of Florida – B.S. in Computer Science, Minor in Digital Arts & Sciences
Dean's List (2022–2023)

August 2018 – August 2023
Gainesville, FL

Coursework: Data Structures & Algorithms, Operating Systems, Software Engineering, Digital Logic, 3D Audio, Database Systems, Computer Networking, Numerical & Data Analysis, Blockchain Development, UI/UX Design, Game Development.